

Predicting Scientific Citations

A Multi-Signal Retrieval & Fusion Pipeline

IR Challenge Final Presentation

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April 16, 2026

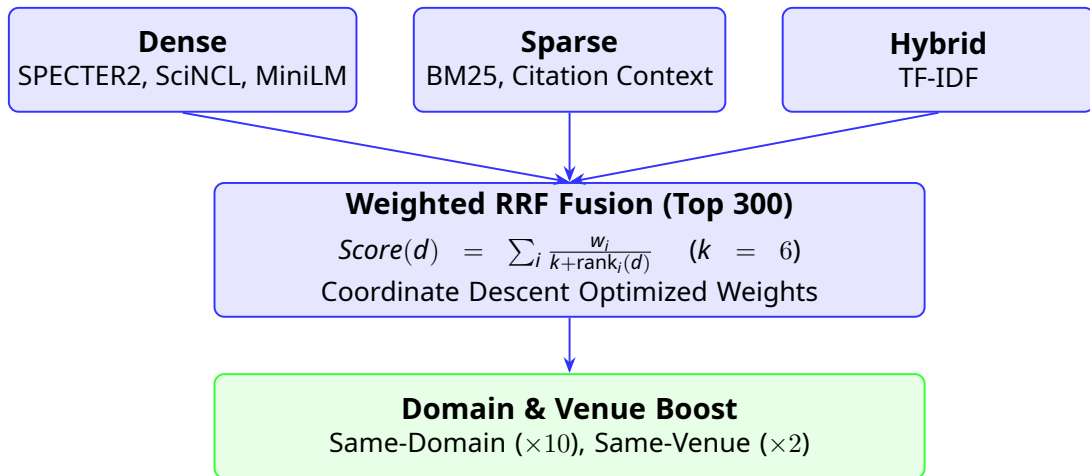
Part 1: Pipeline Overview

Architecture, Design Decisions, and the Core Problem

The Problem: Citation $\neq$ Semantic Search

- **Goal:** Retrieve 100 cited papers from a 20K corpus.
- **Relevance:** Ground truth = actual citations (not just topical similarity).
- **Evaluation:** NDCG@10 (primary), Recall@100, MAP.
- **Key Insight:** Citations are functional (methodologies, datasets), not just semantic.
- **Data Reality:** **97.6%** of citations are same-domain.

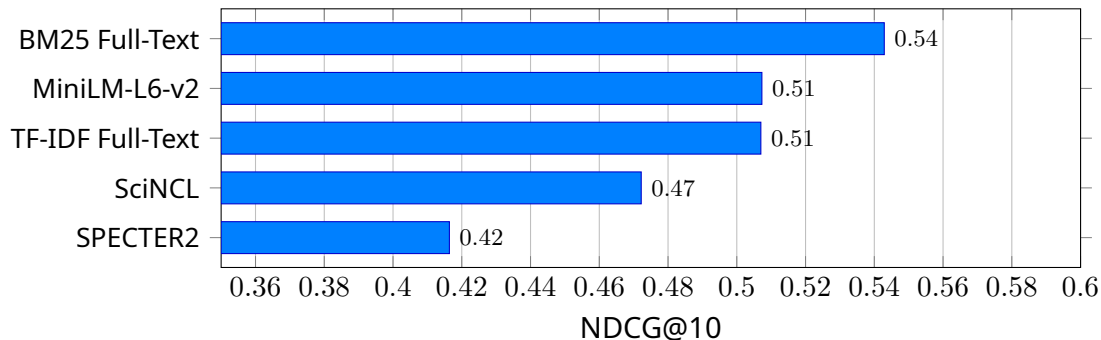
3-Stage Pipeline Architecture



Part 2: Ablation Study

Component Contributions & Signal Effectiveness

Individual Retriever Performance (Public Queries)



Surprise: Lexical (BM25) > Dense Models.

Specific entity matching ("p53", "LSTM") beats generic semantic compression.

Leave-One-Out Component Analysis

Removing components from the baseline v4 fusion (8 signals, NDCG@10 = 0.6151):

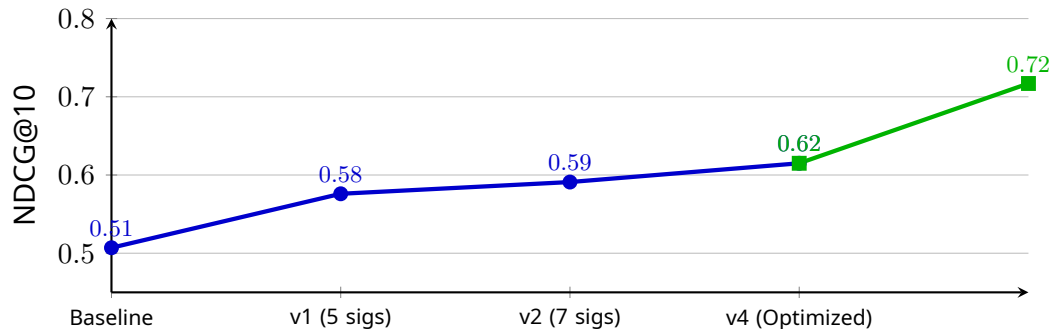
Signal Removed	NDCG Drop	Interpretation
BM25 Full-Text	-0.0540	Most critical — lexical backbone
TF-IDF Full-Text	-0.0295	Second most important
MiniLM-L6-v2	-0.0229	General semantic mapping
SciNCL	-0.0081	Citation graph adds diversity

Takeaway: The strongest ensemble relies on diverse, uncorrelated signals, not just the best standalone models.

Part 3: Results & Progression

Iterative Growth and Leaderboard Context

NDCG@10 Progression (Local Evaluation)



Final Submission Strategy

Submitted with a conservative **10x boost** to mitigate overfitting between local validation and blind held-out tests.

97.6% Same-Domain Citations

- **Impact:** Drove $\sim 50\%$ of total NDCG improvement (+0.10).
- **Why it works:** Neural models fail to encode this hard structural boundary.

Boost Multiplier	NDCG@10
1x (None)	0.6151
2x Boost	0.6925
10x Boost (Submission)	0.7170

Part 4: Error Analysis & Lessons

What Failed, Why it Failed, and Next Steps

Approaches That Failed (And Why)

Cross-Encoder Reranking (-0.115 NDCG)

Failure → **Citation** $\neq$ **Relevance**. Optimizes for broad topics, actively pushing down niche methodology papers.

LLM Reranking (-0.034 NDCG)

Failure → **Wrong Inductive Bias**. LLMs rank by semantic similarity/recency. They lack crucial structural metadata.

Year Filter Constraint (-0.213 NDCG)

Failure → **Check Data Assumptions**. 68.5% of "cited" papers in the dataset are actually *newer* than the query.

Failure Modes & Future Work

Where the Pipeline Struggles

- **The 90% Recall Ceiling:** 9.9% of cited papers are *never* retrieved by any of our 8 signals in the top 300.
- **Interdisciplinary Papers:** Cross-domain citations are penalized by the domain boost.
- **Broad Domains:** "Biology" and "CS" contain massive subfields with near-zero vocabulary overlap.

What We Would Try Next

- **Sub-domain Clustering:** Dynamically split broad domains.
- **Learning-to-Rank:** Replace RRF with LambdaMART using all 8 retrieval scores as features.

Thank you for listening!

Final NDCG@10: 0.6965

Rank: #2

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